

What is the difference between bit code and byte code?

C and C++ language

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Operating System

Bit Code

Java language

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Appropriate JVM

Operating System

Byte code

* Bit code is directly understood by Operating System but as the other hand byte code is understood by JVM, JVM is going to convert the byte code into appropriate machine understandable format.

*** What is the difference between JVM, JRE, JDT and JET compiler?

JDK = JRE + Development tools

JRE = JAW + Class Libraries

JAW = Class Loader + Runtime Data areas + Execution Engine

Interpreter
JIT compiler

JDK :

- It stands for Java Development Kit.
- It is a development bundle that comes by using JDK we can develop as well as execute the java programs.
- As we are to develop and execute the java programs, we have various development tools:
 - javac : java compiler, responsible for compilation.
 - JD : java class loader, responsible for loading java program.
 - JDUI : java debugger, for debugging purpose.
 - JDCompiler : java compiler, to compile the code in the IDE.
 - JDUI : java UI, to get the details of a class.
 - JDUI : java documentation, for generating java documentation.

JRE :

- It stands for Java Runtime Environment. It contains JVM and class libraries.
- It is a client version that means by using JRE we can only execute our java programs (JDK can't execute).
- From java 11 version, JRE folder is removed so now, from java 11+ we can execute a java program without compilation by using following:

```
java PathName.java
```

(This is the command to directly execute our java code)

JAW :

```
public class Java {  
    public static void main(String[] args) {  
        System.out.println("Hello welcome to Java!");  
    }  
}
```

Java Native Java (JAW) can directly execute the program.

JAW :

- It stands for Java Virtual Machine.
- The main process of JVM is to load and execute the .class file.
- JAW takes .class file and convert the .class file into Operating System understandable format.
- Java runtime dependent, it provides support before the class loading, class verification, allocating the memory for all the data members, garbage collection, security manager and so on.

What is the difference between Compiler (JAW) and Interpreter (JAW contains Interpreter) :

Compiler	Interpreter
1) It scans the entire program once.	1) It scans line by line.
2) It converts the .class file into byte code.	2) It converts the byte code into O/S format.
3) Byte code is generated as, it needs a separate memory.	3) Based on the byte code line by line and generate the output concurrently.
4) It displays all the errors and warnings at a time on, debugging is time.	4) It will display only one runtime error at a time as, debugging is time.
5) After successful compilation, execution is fast.	5) It executes the line language and we make a mistake if we make a mistake in the memory.
6) After successful compilation, we can delete the source code.	6) We cannot delete the byte code due to fast by line execution.

Note : Java uses compiler and Interpreter.